Questions 1 – 5

Choose the *correct letter*, A, B, C or D.
Write your answer beside each question number on this paper.

1 Which option best describes the scope of cybersecurity?

- A The legal measures that allow prosecutors to pursue attackers after an incident
- B The technical measures that block unauthorised access to networks and applications
- C The managerial measures that document security policies and staff responsibilities
- D The technical, legal, managerial and social measures that protect the cyber environment

2 Which set of developments belongs to the period from the 2010s onwards?

- A ARPANET together with the very earliest mainframe installations
- B Advanced persistent threats, ransomware, IoT, AI and zero trust models
- C The first computer viruses, worms and commercial antivirus products
- D The integration of cybersecurity into e-commerce and e-banking services

3 Which responsibility for cybersecurity falls specifically to governments?

- A Protecting cross-border digital commerce and international payment systems
- B Protecting customer trust, business continuity and legal compliance
- C Protecting national security, defence and e-governance infrastructure
- D Protecting personal emails, social media accounts and banking details

4 Which pair of controls is used to secure IoT systems?

- A Lightweight encryption together with anomaly detection
- B Firewalls together with intrusion detection and prevention systems
- C Secure coding together with the OWASP standards
- D Sandboxing together with malware detection

5 Which set of techniques is used to enforce confidentiality?

- A Redundancy, load balancing and DDoS mitigation
- B Encryption, access controls and virtual private networks
- C Sandboxing, malware detection and application isolation
- D Hashing, digital signatures and version control

Questions 6 – 10

Choose the *correct letter*, A, B, C or D.
Write your answer beside each question number on this paper.

6 A university registration portal becomes unreachable during the enrolment peak and no student can log in. Which pillar has failed, and which controls best prevent a repeat?

A Integrity, prevented by hashing, digital signatures and version control
B Availability, prevented by redundancy, load balancing and DDoS mitigation
C Confidentiality, prevented by encryption, access controls and VPNs
D Non-repudiation, prevented by signed audit logs and trusted timestamps

7 An analyst estimates the probability of each threat and its impact for the coming year. Which expression gives the expected cyber risk exposure at a point in time?

A The sum of each threat probability divided by its impact value
B The ratio of the total impact to the number of identified threats
C The product of the highest probability and the highest impact
D The sum of each threat probability multiplied by its impact value

8 A hospital assigns a higher weight to confidentiality than to availability when scoring a system. In the utility function $U = \alpha C + \beta I + \gamma A$, what do the weights represent?

A The cost of the controls that the organisation deploys for each pillar
B The number of incidents recorded against each pillar during one year
C The importance that the given system assigns to each of the three pillars
D The measured performance of each pillar, normalised between zero and one

9 The attack surface metric multiplies three quantities for every component of a system. Which three quantities does it use?

A External exploits, validated countermeasures and assigned owners
B Encrypted endpoints, verified identities and applied access policies
C Exposed entry points, vulnerability severity score and asset value
D Estimated exposure, vendor rating and annual audit frequency

10 An auditor measures how far an organisation satisfies each policy requirement and how important each requirement is. Which expression gives the weighted policy compliance index?

A The weighted sum of compliance levels divided by the total of the weights
B The highest compliance level divided by the total of the weights across requirements
C The arithmetic mean of compliance levels divided by the total of the weights
D The weighted sum of compliance levels divided by the number of requirements

Questions 11 – 15

Choose the *correct letter*, A, B, C or D.
Write your answer beside each question number on this paper.

11 One sector places a higher weight on confidentiality than on availability, while another sector ranks the pillars differently. Which reasoning best explains why such weights differ between sectors?

A The weights are chosen at random to keep the utility function mathematically stable
B The weights reflect the relative harm caused when each pillar is violated in that sector
C The weights follow the alphabetical order of the pillars inside the utility function
D The weights are fixed constants that apply identically to every sector and system

12 Global ransomware damage is expected to exceed twenty billion dollars, and cloud-related attacks are rising. Which conclusion is best supported by these trends?

A Ransomware has become technically impossible to detect on modern computer networks
B Attackers have abandoned malware in favour of purely physical intrusion methods
C Economic damage and misconfiguration-driven exposure are both increasing over time
D Cloud providers have eliminated the misconfigurations that caused earlier breaches

13 Advanced persistent threats are state-sponsored and stealthy long-term attacks. How do they differ from ordinary malware infections?

A They rely solely on physical theft of media and never use network access
B They remain inside the target for a long period and avoid immediate detection
C They target only home users and never reach enterprise or government networks
D They spread automatically to every host and delete all data within minutes

14 Which sequence states the meanings of risk, vulnerability and exploit correctly?

A Risk as an attack tool, vulnerability as likelihood and impact, exploit as a weakness
B Risk as a weakness, vulnerability as an attack tool, exploit as likelihood and impact
C Risk as a likelihood only, vulnerability as an impact only, exploit as a severity score
D Risk as likelihood and impact, vulnerability as a weakness, exploit as an attack tool

15 Stuxnet and the Colonial Pipeline incident are two widely studied cyber incidents. What does this pairing illustrate about cybersecurity?

A Both show that isolated industrial systems can never be reached by any attacker
B Both show that attacks remain confined to digital data with no physical effects
C Both show that attacks can disrupt physical operations and critical infrastructure
D Both show that only financial institutions are targeted by advanced attackers

Questions 16 – 20

Choose the *correct letter*, A, B, C or D.
Write your answer beside each question number on this paper.

16 Which pairing of regulation and protected domain is correct?

A GDPR covers European data protection, while HIPAA covers healthcare information
B GDPR covers healthcare information, while HIPAA covers European data protection
C GDPR covers international standards, while HIPAA covers European data protection
D GDPR covers European data protection, while HIPAA covers international standards

17 Cybersecurity can be modelled as an optimisation problem whose objective is to minimise risk while maximising security investment. Which statement best evaluates that objective?

A Risk rises as investment rises, because extra controls create additional attack paths
B Risk is minimised by ignoring threat probability and paying for every available control
C Risk is independent of both the investment level and the threat probabilities
D Risk falls as investment rises, but the model treats control cost as a linear deduction

18 An organisation protects laptops, a public cloud tenant, an IoT sensor fleet and a factory control network. Which conclusion about domain-specific controls is best supported?

A Each domain needs controls matched to it, so one universal control set is insufficient
B Only the factory control network matters, since the other domains carry no real risk
C One control set is enough, because firewalls protect every domain equally well
D Only the public cloud tenant matters, since laptops and sensors store no data

19 Zero trust has become a leading security model in the last decade. Which statement best evaluates why perimeter-only defence became insufficient?

A Users, devices and services now sit outside a single well-defined network perimeter
B Employees stopped using passwords and moved entirely to hardware security keys
C Firewalls became technically unable to filter traffic on modern high-speed links
D Attackers stopped targeting networks and moved entirely to physical intrusions

20 A team must reduce the attack surface of a system. Which decision is the most defensible?

A Keep every entry point open and rely on the asset values remaining unchanged
B Retire all high-value assets so that the metric can no longer be computed
C Open additional entry points so that attackers become confused by the extra services
D Close unnecessary entry points and patch severe vulnerabilities on high-value assets

Answer key

1	D	6	B	11	B	16	A
2	B	7	D	12	C	17	D
3	C	8	C	13	B	18	A
4	A	9	C	14	D	19	A
5	B	10	A	15	C	20	D